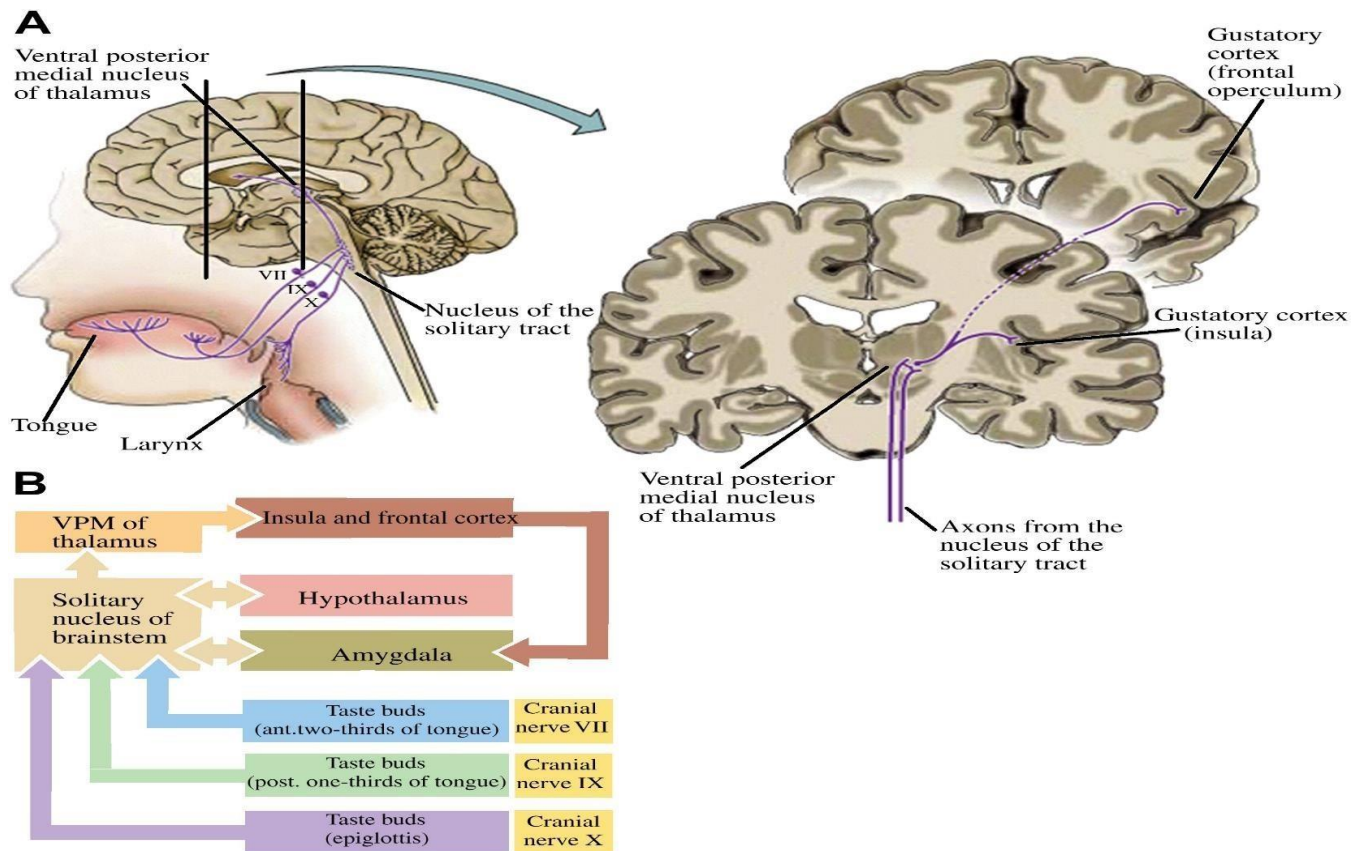


Lecture (20): Gustatory and Olfactory Pathways

Gustatory pathway:

- **Gustatory information** is detected by **chemoreceptors on taste buds** found within:
 - **Lingual papillae** on the dorsum of the tongue.
 - The **epithelium lining** parts of the larynx, pharynx and epiglottis.
- There are approximately **4000 taste buds**.
- **Taste buds** are formed by groupings of **taste receptor cells**.
- Neural impulses from the taste receptors are transferred to the brain through **three gustatory nerves**:
 - 1- **Chorda tympani** of the **facial nerve (CN VII)** carry taste from the **anterior two-thirds of the tongue**. The chorda tympani is formed by peripheral processes of cells of **Geniculate ganglion of facial nerve (1st order neuron)**. after which the efferent fibers form the **nervus intermedius** (sensory root of facial).
 - 2- **Lingual branch** of the **glossopharyngeal nerve (CN IX)** carries taste from the posterior third of the tongue to inferior ganglion of CN IX (**1st order neuron**).
 - 3- **Internal laryngeal branch** of the **vagus nerve (CN X)** carries taste sensation from the **root of the tongue and the epiglottis to the inferior vagal ganglion (1st order neuron)**.
- **2nd order neuron**: the **nucleus solitarius** of the **medulla oblongata** receives gustatory afferents of the facial, glossopharyngeal and vagus nerve.
 - Most of the fibers of nucleus **solitarius** ascend as "**Solitario-thalamic tract**" and synapse in the ventral posterior medial nucleus of the thalamus (**PMVNT**).
 - Other fibers from **nucleus solitarius** travel to:
 - 1- **Lateral hypothalamic area**, which is involved in appetite and satiety.
 - 2- **Amygdala**, which is involved in emotions reactions to food.
 - 3- **Superior and inferior salivatory parasympathetic nuclei**, which supply the salivary glands to promote secretion of saliva.
- **3rd order neuron**: Cell bodies of the **ventral posterior medial nucleus** of the **thalamus (PMVNT)**.
 - The axons of (**PMVNT**) project to the **gustatory cortex, area 43** (upper part of **insula & frontoparietal operculum**).



Olfactory Pathway:

- **Olfactory receptor cells:**

They are **bipolar cells** located in the **olfactory epithelium** that lines the roof and sides of the nasal cavity.

- **Olfactory nerve (CN I):** formed of bundles of nerve fibers of the olfactory receptor cell axons, passing superiorly through the foramina in the **cribriform plate of the ethmoid bone**.

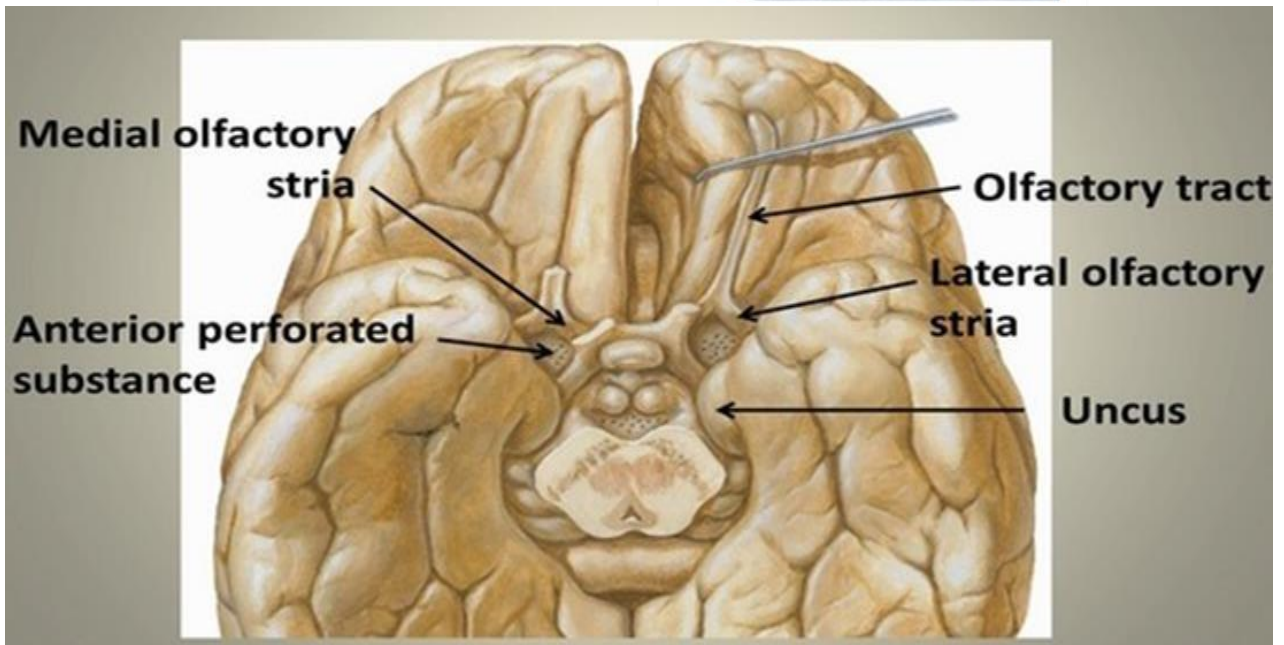
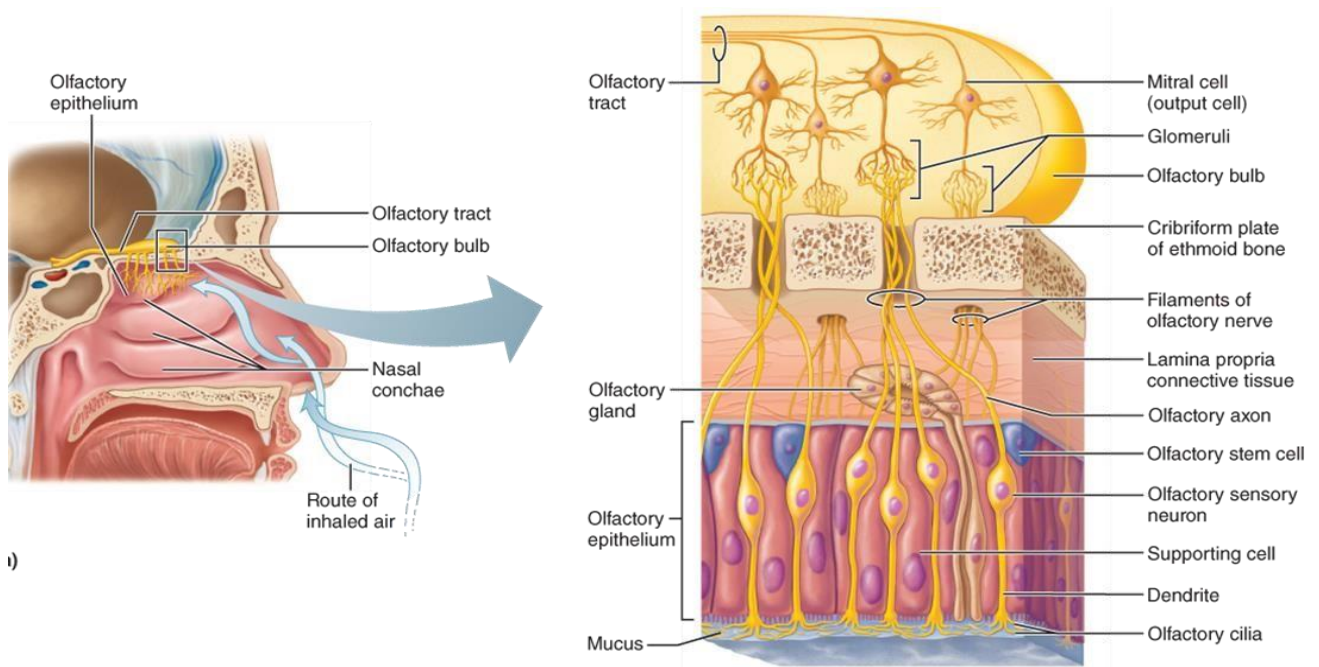
- **Olfactory bulb:**

- Each olfactory bulb lies superior to the cribriform plate of the ethmoid bone, inside the cranial cavity.
- It is the relay station of the olfactory pathway and contains olfactory glomeruli.
- Contains the **2nd order neuron (Mitral & Tufted cells)**.

- **Olfactory tract:**

- It is made up of **axons of Mitral & Tufted cells** of olfactory bulbs.
- The tract passes posteriorly on the underside of the medial frontal lobe in **olfactory groove**.
- The tract on both sides is divided into **medial and lateral olfactory striae**.

- The **medial olfactory stria**: projects to the anterior commissure to connect with the contralateral olfactory bulb.
- The **lateral olfactory stria**: terminate on the **olfactory cortex**.
- **Primary olfactory cortex**:
 - **Piriform cortex**: which is located below the lateral olfactory stria.
 - **Entorhinal cortex**: which is the anterior part of **parahippocampal gyrus**.
- **Secondary olfactory cortex**:
 - **Orbitofrontal cortex** to **distinguish different odors**. ➤ **Amygdala** involved in **emotional reactions** to odors.
 - **Hypothalamus** involved in **autonomic responses** (appetite, salivation, and



gastric contraction). **Hippocampus** for **olfactory** memories. **Insular cortex**.